

Payslip Generator Project Documentation

Project Title

Payslip Generator Web Application

Project Overview

The Payslip Generator is a web-based application developed to automate employee payslip generation. The application calculates employee earnings, deductions, taxes, paid days, and net salary automatically based on user input.

The project also supports employee data storage and retrieval using MySQL database integration.

Technologies Used

Frontend

- HTML5
- CSS3
- Bootstrap 5
- JavaScript

Backend

- Node.js
- Express.js

Database

- MySQL

Additional Libraries

- Flatpickr (Date Picker)
 - Fetch API
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Project Features

Employee Management

- Add employee details
- Search employee by name or employee ID
- Save employee data into MySQL database

Payslip Features

- Automatic Total Days Calculation
- Automatic Paid Days Calculation
- Net Pay Calculation
- HRA Calculation
- PF Contribution Calculation
- Income Tax Calculation
- Total Earnings Calculation
- Total Deductions Calculation
- Year-To-Date (YTD) Calculation

Validations

- PAN Card Validation
- Numeric Value Validation
- Auto Uppercase Formatting

UI Features

- Responsive Design using Bootstrap
- Dynamic Payslip Display
- Auto-updating salary calculations
- Calendar Date Picker

Folder Structure

```
Payslip_Project
|
├── public
|   ├── index.html
|   ├── style.css
|   └── script.js
|
└── server.js
```

```
├─ db.js
├─ package.json
└─ node_modules
```

Database Configuration

Database Name

payslip_db

Example Table

```
CREATE TABLE employees (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(100),  
  personal_id_no VARCHAR(50),  
  designation VARCHAR(100),  
  pan VARCHAR(20),  
  department VARCHAR(100),  
  pay_from VARCHAR(20),  
  pay_to VARCHAR(20),  
  paid_days VARCHAR(20),  
  uan VARCHAR(50),  
  pf_no VARCHAR(50),  
  bank_name VARCHAR(100),  
  bank_account VARCHAR(100),  
  basic_pay DECIMAL(10,2),  
  hra DECIMAL(10,2),  
  transport DECIMAL(10,2),  
  meal DECIMAL(10,2),  
  total_earnings DECIMAL(10,2),  
  income_tax DECIMAL(10,2),  
  pf_contribution DECIMAL(10,2),  
  prof_tax DECIMAL(10,2),  
  medical DECIMAL(10,2),  
  sodexo DECIMAL(10,2),  
  total_deductions DECIMAL(10,2),  
  net_pay DECIMAL(10,2)  
);
```

Salary Calculation Logic

Paid Days Formula

$\text{Paid Days} = \text{Total Days} - (\text{Before Joining Leaves} + \text{Leave Days})$

HRA Formula

$\text{HRA} = \text{Prorated Basic Pay} \times 50\%$

PF Contribution Formula

$\text{PF Contribution} = \text{Prorated Basic Pay} \times 12\%$

Income Tax Formula

$\text{Income Tax} = \text{Total Earnings} \times 10\%$

Net Pay Formula

$\text{Net Pay} = \text{Total Earnings} - (\text{Total Tax} + \text{Total Deductions})$

Steps To Run Project

Step 1

Install Node.js and MySQL.

Step 2

Open project folder in VS Code.

Step 3

Install dependencies.

```
npm install
```

Step 4

Start MySQL server.

Step 5

Run backend server.

```
node server.js
```

Step 6

Open browser.

```
http://localhost:3000
```

Challenges Faced

- MySQL connection issues
- Port conflicts
- Paid Days auto-calculation errors
- Calendar picker issues
- Duplicate deduction calculations
- Frontend and backend integration issues

Learning Outcomes

- Learned frontend and backend integration
 - Learned MySQL database connectivity
 - Learned salary calculation logic
 - Learned API integration using Fetch API
 - Learned validation techniques in JavaScript
 - Learned responsive UI design using Bootstrap
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Future Enhancements

- PDF Payslip Download
 - Email Payslip Feature
 - Admin Login System
 - Employee Dashboard
 - Excel Export
 - Authentication and Security
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Conclusion

The Payslip Generator project successfully automates payroll calculations and employee payslip management. The project improved understanding of full-stack web development, database management, salary calculation logic, and real-time form handling.